

The Thrower's Shoulder

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The thrower's shoulder is a unique and challenging subset of sports medicine given the extreme demands placed on the glenohumeral joint and surrounding soft tissue structures. For purposes of discussion, we will mainly focus on medical issues as they relate to baseball players; however, we certainly acknowledge that many other sports (volleyball, tennis, handball) and positions (quarterbacks) have overlapping demands and injury patterns and will draw on literature based in those disciplines as well.

Generally, these are complex conditions that can present several difficulties in both diagnosis and treatment. The ability to throw a baseball at velocities exceeding 90 miles/h is the result of a complex transfer of energy from the lower extremity through the trunk, shoulder, elbow, and hand. The stresses placed on the individual links of the kinetic chain during throwing can lead to failure of and injury to these structures. The repetitive acceleration and deceleration of the arm during the throwing motion subjects the shoulder to extreme positions and stresses, which can lead to chronic overuse-type as well as acute shoulder injuries.

The thrower's shoulder endures supraphysiologic forces as professional pitchers have demonstrated the ability to generate up to 92 Nm of humeral rotation torque, which is greater than the torsional failure limit found in cadavers.¹ The shoulder experiences forces nearly half of body weight during the late cocking phase and distraction forces of nearly the entire body weight during the deceleration phase with peak angular velocities of 7250 degrees/s.^{2,3} To perform such supraphysiologic feats, the shoulder makes key adaptations over time, which will be discussed further. However, the magnitude and repetition of such stresses placed on the shoulder can lead to injuries to important structures.

Sports medicine physicians who treat overhead athletes must be familiar with the subtle changes in their performance to recognize an injury to the shoulder. Differing from the general population, pitchers may not report pain or weakness, but instead complain of decreased velocity or accuracy. A thorough understanding of the anatomy and adaptive changes that occur in the thrower's shoulder along with the physiology of throwing is necessary to provide treatment in these complicated cases. In addition, a systematic approach to evaluating the shoulder is critical as many injuries are multifactorial due to mechanical, technical, training-related, and structural factors.

ADAPTATIONS TO THE THROWING SHOULDER

Over time, overhead athletes develop well-described adaptations in response to the stresses placed on the shoulder as a result of throwing. These changes likely begin in early childhood as the athlete begins throwing regularly. Both soft tissue and bony adaptations occur, and while sometimes subtle, they are imperative to the thrower's ability to function at a high level under otherwise nonphysiologic conditions.

One of the primary physiologic changes in the thrower's shoulder is seen in range of motion (ROM). Most throwers exhibit a difference in the amount of external rotation (ER) and internal rotation (IR) between their throwing and non-throwing shoulders, generally with an increase in ER and decrease in IR on their dominant side (Fig. 45.1).² A comparative loss of IR greater than 20 to 25 degrees has been termed glenohumeral internal rotation deficit (GIRD) although the pathologic nature of this change has been called into question more recently.

Brown et al.⁴ reported that professional pitchers had a mean shoulder ER of 141 degrees with the arm at 90 degrees of abduction, which was 9 degrees more than their nondominant arm and also 9 degrees more than the throwing shoulders of position players. Although the magnitude of IR and ER seen in the dominant and nondominant arm differ, the total arc of motion is usually similar. Wilk et al.⁵ reported a mean ROM arc of 176 degrees (both sides) and that the total motion in the throwing shoulder was within 7 degrees of the contralateral side. GIRD has classically been felt to be a maladaptive change, and indeed, some modern literature does support increased injury risk with GIRD.^{6,7} However, total arc of motion⁶⁻⁸ and even ER deficits⁹ are increasingly recognized as perhaps more problematic and significant risk factors for injury.

Certainly, soft tissue accommodations occur over time in the static and dynamic stabilizers surrounding the shoulder, but bony adaptations have also been well documented. Several authors have demonstrated a 10- to 17-degree increase in humeral retroversion in the dominant arms of throwers relative to the contralateral side, which allows for increased external rotation.¹⁰⁻¹² Yamamoto et al.¹³ assert that the repetitive throwing motion does not directly increase the retroversion of the humeral head but rather restricts the natural, physiologic derotation of the humeral head during growth. And while it is unclear at what

Abstract

The thrower's shoulder is a unique and challenging subset of sports medicine given the extreme demands placed on the glenohumeral joint and surrounding soft tissue structures. The uniqueness of this population is based on the complex energy transfer starting in the lower extremities and propagating up through the trunk, shoulder, elbow and hand. Specifically as it relates to the shoulder, tremendous neuromuscular control is required to appropriately position the scapula, and by extension, the glenohumeral joint, into the correct position throughout this dynamic process. Throwers with a painful shoulder should thus be addressed individually with a focused history and physical exam and special emphasis on the nature of the pain and where it occurs during the throwing motion. Imaging is often beneficial, particularly magnetic resonance imaging scans, and can help to delineate the state of the rotator cuff, cartilage, and labrum. When approaching treatment, the initial management of the majority of overhead athletes is nonoperative—consisting of a period of rest, activity modification and rehabilitation. We feel that the importance of high quality physical therapy with an overhead throwing specialist cannot be overstated. Whether the pathology involves internal impingement, partial-thickness rotator cuff tears, superior labral from anterior to posterior (SLAP) lesions, biceps pathology, or glenohumeral internal rotation deficit, we recommend proceeding with caution when addressing these issues surgically as the results have historically been sobering. Overall, overhead athletes are a high-demand but rewarding patient population, and physicians should approach their care with a solid knowledge base, realistic expectations, and stringent surgical criteria.

Keywords

throwing shoulder
scapular dyskinesis
glenohumeral internal rotation deficit (GIRD)
internal impingement
SLAP tear

age these bony changes begin to occur, they certainly appear to represent adaptive growth over time.

Soft tissue adaptations are also seen as a result of the large rotational and distractive forces acting upon the glenohumeral joint, resulting in posterior capsular tightness and anterior capsular laxity.^{14–17} The term microinstability has been used to describe the acquired capsular laxity that allows increased humeral head translation and rotation. While a variety of theories exist, the cause of this microinstability is believed to be repetitive tensile loading as a result of scapular protraction or repetitive ER, particularly as it relates to the deceleration phase of throwing.^{15–18} But despite the restrictive posterior soft tissue envelope, pitchers appear to have increased overall laxity of the shoulder compared with athletes who play other positions. Bigliani et al.¹⁴ found a sulcus sign in 61% of dominant shoulders in professional pitchers compared to only 47% in position players. While there is likely a congenital component that self-selects for shoulders able to tolerate repetitive overhead motion, these capsular changes appear to be a necessary (as opposed to pathologic) adaptation to throwing as increased anterior–posterior humeral head translation has been demonstrated in both symptomatic and asymptomatic pitchers.^{15,19}

BIOMECHANICS OF THROWING

To fully understand the spectrum of injuries, the stresses placed on the shoulder during the action of throwing must be



Fig. 45.1 An overhead throwing athlete with asymmetric loss of internal rotation. The throwing shoulder shows an increase in external rotation, leading the average arc of motion of the arms to be similar. (From Burkhart SS, Morgan CD, Kibler WB. The disabled throwing shoulder: spectrum of pathology I: pathoanatomy and biomechanics. *Arthroscopy*. 2003;19:404–420.)

understood. The overhead throwing motion can be broken down into six discrete phases: wind-up, early cocking, late cocking, acceleration, deceleration, and follow-through (Fig. 45.2).^{2,17,20,21} The entire motion takes approximately 2 seconds although 75% of that time occurs during the pre-acceleration phases.^{21–23}

During the first phase of throwing (wind-up), the body rises over the center of gravity and the shoulder is placed in slight abduction and IR. At this point in the pitching motion, no stress is placed on the upper extremity.^{24–26}

Early cocking is the second phase where the arm is placed into abduction and ER, and the arm rotates behind the body axis approximately 15 degrees. Once the arm reaches the top of its motion and stops moving backward, this is the onset of the late cocking phase.

The third phase is late cocking, which begins as the lead leg contacts the ground and ends when the arm reaches maximal ER of nearly 180 degrees. During this phase, the scapula retracts to provide a stable glenoid surface for the humeral head to compress against. The upper arm is maintained in 90 degrees to 100 degrees of abduction, and the elbow moves even with the plane of the torso. The humerus progresses into ER, and the humeral head translates posteriorly on the glenoid because of the increasing tightness of the anterior structures. The infraspinatus and teres minor muscles are active early in the late cocking phase, leading to the ER of the humerus. The subscapularis becomes active late in this phase as IR begins. The result is that the rotator cuff as a whole compresses the humeral head against the glenoid with significant force.²

The acceleration phase begins as the humerus internally rotates and ends when the ball is released. Although significant angular velocity is developed by the muscular forces around the shoulder during this phase, little stress is noted in the shoulder.²⁷ The triceps becomes active early in the acceleration phase, followed by the pectoralis major and the latissimus dorsi muscles later.²⁸

The fifth phase (deceleration) begins just after the ball is released and ends when humeral IR ceases. This phase leads to tremendous stresses generated by the rotator cuff muscles as the arm is brought to an abrupt halt. Eccentric loads on the posterior cuff reach 1000 N as the muscle dissipates the kinetic energy generated in the earlier phases of the throw.^{28,29}

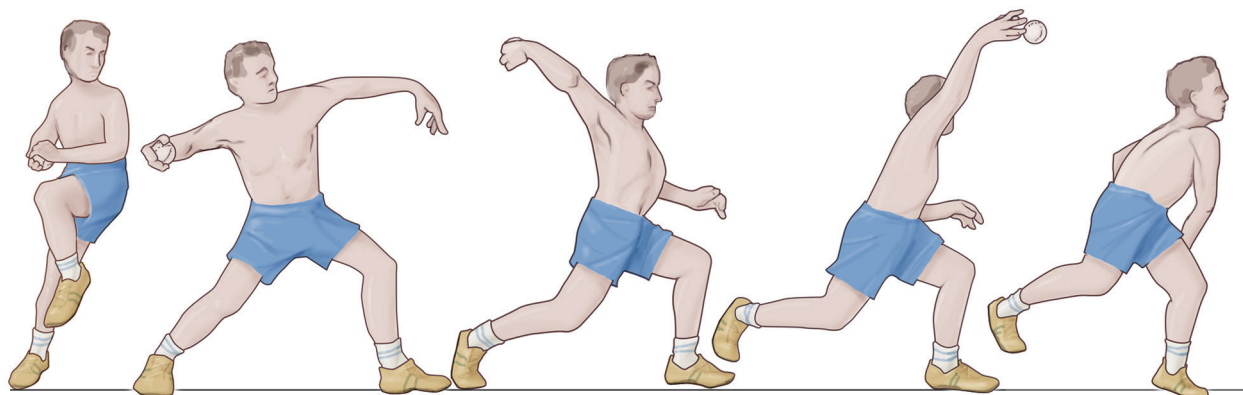


Fig. 45.2 Stages of the throwing motion.

The final phase of the throwing motion is the follow-through. It is at this point of the pitch that the body regains balance and stability. The muscles cease to fire and the compressive forces across the glenohumeral joint drop significantly.

PATHOPHYSIOLOGY

Internal Impingement

In 1992, Walch et al.³⁰ initially reported on contact that occurs between the supraspinatus tendon and the posterosuperior edge of the glenoid cavity in tennis players and coined the term internal impingement. It represents a collection of injuries characterized by pathologic contact between the rotator cuff (posterior supraspinatus and anterior infraspinatus) and the greater tuberosity with the posterosuperior glenoid and labrum leading to articular-sided rotator cuff tears and posterosuperior labral tears (Fig. 45.3).³¹ This contact is only possible by placing the shoulder in the supraphysiologic position of extreme ER and anterior translation seen while throwing. While the concept of internal impingement endures, no clear etiology has been established, and multiple theories exist.

The theory of microinstability centers on the idea of hyperangulation—that as the humeral shaft moves posteriorly out of the plane of the scapula, the accompanying weakness allows the humeral head to drift anteriorly and subsequently produces a lengthening of the anterior capsule, which leads to a subtle form of anterior instability.³² Jobe and Pink believed that the repetitive and forceful overhead activity causes a gradual stretching out of anteroinferior capsuloligamentous structures, leading to subclinical instability and impingement.³³ Indeed, throwers often have an element of hyperlaxity as previously discussed, and some would posit that tipping the scale from laxity to instability is the catalyst for developing internal impingement.³⁴

Burkhart and Morgan³² proposed that the pathology is a result of a hypertwisting mechanism with large shear stresses leading to fatigue failure of both the rotator cuff and the biceps tendon insertion point of the labrum. These lesions result from an acquired posterosuperior instability that is caused by posteroinferior

capsular tightness, which they termed the peel-back mechanism. They reasoned that the posterior capsule must withstand tensile forces of up to 750 N during the deceleration and follow-through phases of throwing. The posterior tensile forces placed on the shoulder during these actions are offset by the eccentric contraction of infraspinatus and the posteroinferior capsule, including the posterior band of the inferior glenohumeral ligament. The posterior contraction is believed to shift the center of rotation of the shoulder to a more posterosuperior location, creating instability with the shoulder in abduction and ER.

Muscle fatigue and imbalance may also play a role in the development of internal impingement by altering the mechanics of the shoulder. Fatigue-related humeral hyperextension can occur during the late cocking phase when the rotator cuff muscles cannot completely resist the large acceleration forces generated while pitching.³⁵ Progressive delamination of the posterior capsulolabral structures and the rotator cuff may occur as the violent deceleration of the arm during follow-through may cause an abrasive degeneration of the rotator cuff on the posterosuperior aspect of the glenoid.³⁶ Whether it is the driving force or simply a result of the process, scapular dyskinesis is also often seen with internal impingement.³⁷

Although a range of theories have been proposed regarding the pathophysiologic etiology of internal impingement, it seems clear that the causes of disability in the shoulder used for throwing are multifactorial and require an equally multicentric approach to care.

Dynamic Shoulder and Scapular Stability

Particularly in overhead athletes, focal weakness and muscle imbalance play a role in the development of shoulder pain. Multiple studies have demonstrated the protective effect of maintaining shoulder strength (especially ER strength) as a means of preventing injuries.^{6,8,38} Any muscle imbalance alters the anterior–posterior force couple that stabilizes the glenohumeral joint and increases the compression forces in the joint.^{39,40} Disrupting the agonist–antagonist balance impairs a pitcher's ability to appropriately accelerate and decelerate while throwing.

Although often neglected, the scapula is the critical link in the kinetic chain of energy transfer from the trunk to the humerus.^{41,42} The scapula provides a stable base upon which the rotator cuff muscles activate, and it coordinates scapulohumeral rhythm, the coupled movement between the scapula and the arm allowing for placement of the upper extremity in space. Scapular movement combines three motions (upward–downward rotation, internal–external rotation, and anterior–posterior tilt) with two sliding translations (up–down and anterior–posterior). It is commonly believed that scapular malposition can also be a cause of shoulder problems in throwers.^{27,43}

Kibler and Thomas⁴⁴ described dyskinesis as an alteration of static scapular position or dynamic scapular motion in coordination with arm motion. The resulting protracted, downward rotated and anteriorly tilted scapula is caused by abnormal muscular imbalances.^{45,46} The protracted scapula is almost universally unfavorable for shoulder function as it decreases the subacromial space and effectiveness of the rotator cuff while also increasing impingement and strain on the static glenohumeral ligaments

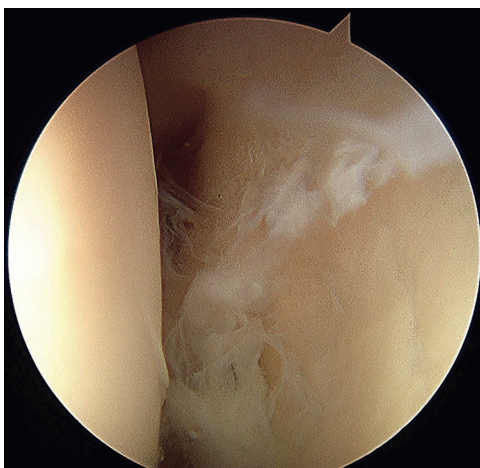


Fig. 45.3 An arthroscopic view of the superior labrum contacting the posterior aspect of the supraspinatus. These findings are consistent with internal impingement.

and dynamic shoulder stabilizers.^{41,47} Accordingly, most goals of management center on regaining functional retraction. And while dyskinesia may itself be the source of a thrower's dysfunction, it can also merely be the byproduct of underlying pathology. Recognition by the physician is key, but high-quality rehabilitation by a thoughtful physical therapist is the cornerstone of treatment for dyskinesia with the possibility of operative intervention if concomitant surgical pathology exists.

The mnemonic SICK scapula was originally coined by Burkhart and colleagues in 2003 and is a subset of dyskinesia relating exclusively to overhead athletes.¹⁵ It represents scapular malposition, inferior medial border prominence, coracoid pain, and dyskinesia. The hallmark feature is a "dropped" scapula with asymmetry in the throwing shoulder where the scapula sits lower than the nondominant side. This static asymmetry is compounded by a dynamic dyskinesia as the athlete goes through the throwing motion. The physical exam should consist of static scapular measurements with the patient standing, a meticulous coracoid exam (as the anterior pain can mimic labral pathology), and pain-free forward flexion while performing the scapular retraction test. These changes in scapular position lead to external impingement as a result of anterior tilt, internal impingement, decreased rotator cuff strength, and anterior capsular strain.^{2,3,18,31,46,48–52} Altered static and dynamic scapular mechanics arise from overuse and weakness of scapular stabilizers and posterior rotator cuff muscles.⁵³ Thus SICK scapula is a nonoperative condition that improves with diligent rehabilitation; however, success hinges on athlete compliance and the ability to get the scapula back to a symmetric position with the contralateral side and keep it there.¹⁵

Glenohumeral Internal Rotation Deficit

Bony and soft tissue adaptations over time allow pitchers to develop a relative increase in ER and loss of IR compared to their nondominant side as well as the general population. As discussed, GIRD has classically been felt to be a maladaptive state with an increased injury risk, but this has been called into question more recently. Posterior capsule tightness has been shown to result in multiple kinematic alterations such as increased ER but decreased IR, horizontal adduction, and flexion.^{54,55} It has been associated with multiple injuries including external (subacromial) impingement, internal impingement, and superior labral from anterior to posterior (SLAP) tears.^{6,7,16,28,37,56} More recently, the importance of total arc of motion as opposed to GIRD^{6–8} and the importance of ER deficits over IR deficits⁹ as it relates to injury risk has been established. Further investigation may help to parse out whether GIRD is truly pathologic or merely a necessary adaptation to repetitive throwing.

HISTORY

The etiology of a shoulder injury in an overhead throwing athlete is best determined by taking a thorough history, and thus a familiarity with not only the sport but also position-specific demands is required. Pitchers, for instance, frequently describe feelings of heaviness/sluggishness, stiffness, fatigue, weakness, and the inability to "bring it."⁴⁴ These subjective complaints are often accompanied by objective findings of decreased velocity

on fastballs, lack of movement on breaking balls, and decreased accuracy. While acute injuries do occur in overhead throwing athletes, it is much more common for injuries to be a result of overuse and fatigue. The timing of the onset of symptoms, current and previous treatment modalities, and a history of a previous injury to the shoulder should be determined. The athlete should also be asked specific pitching-related questions including any recent change in mechanics, development of a new pitch, increased pitch count, alteration to training regimen, in what phase of the throwing the pain occurs, and if any other part of the body has been injured that could be changing the throwing mechanics. The role of the hip, core, and lumbar spine are also critical as injuries to these areas can lead to compensatory alterations in throwing mechanics and increased stress on the shoulder.

Typical symptoms in an overhead throwing athlete with pain include anterosuperior or posterosuperior shoulder pain in the late cocking phase. Labral tears specifically may lead to mechanical symptoms such as popping, locking, or snapping. Although the details are beyond the scope of this chapter, questioning the athlete about symptoms related to the cervical spine or thoracic outlet syndrome is also critical to rule out pathology masquerading as shoulder pain.

PHYSICAL EXAMINATION

A thorough and systematic physical examination should be performed for all throwers who present with shoulder pain. The aid of a physical therapist and/or athletic trainer with a solid understanding of throwing mechanics can also be helpful to isolate dynamic issues less familiar to clinicians. The examination should focus not just on the upper extremity but also on the entire kinetic chain, including the lower extremities and trunk. The athlete's natural posture may be evaluated while he or she walks into the room and during history taking. The overall alignment, shoulder height and position, pelvic tilt and rotation, lower extremity alignment, head and neck position, and arm position must be evaluated. Functional movements should be assessed to determine hip and trunk control, muscle imbalance, and inflexibilities, which can be accomplished by having the patient balance on one leg and do a single leg squat. While performing these maneuvers, the patient should be observed for compensatory movements such as pelvic tilt or rotation. Ankle flexibility, lumbar and thoracic spine, and shoulder girdle mobility can be assessed by having the patient perform a full squat with heels on the ground and arms overhead. Flexibility of the hamstrings and the thoracic and lumbar spine can be assessed by having the patient perform a straight-leg toe touch. While supine, both hips should be evaluated for ROM and to determine if any femoroacetabular impingement is present.

Once a comprehensive examination of the remainder of the kinetic chain is completed, a focused evaluation of the affected shoulder is performed. The entire shoulder complex must be visualized. Male patients are asked to remove their shirts, and female athletes should be asked to wear a tank top or sports bra. First, the shoulder is inspected visually to evaluate for muscular atrophy or scapular winging. The position of the scapula at rest should be noted, including asymmetry of scapular tilt, rotation,

and elevation or depression. After visual inspection, the shoulder should be systematically palpated to identify areas of pain. All bony prominences are palpated, including the bicipital groove, greater tuberosity, coracoid, and acromioclavicular (AC) joint.

Shoulder ROM, both active and passive, should be assessed in both standing and supine positions with special attention given to glenohumeral and scapulothoracic motion. Glenohumeral joint passive ROM is assessed for ER and IR at 90 degrees of abduction and for ER at 45 degrees of abduction in the scapular plane. Increased ER with the arm at the side with the patient in the supine position can be reflective of rotator interval laxity. Increased ER in the abducted and externally rotated position reflects anteroinferior capsular laxity and/or humeral retroversion. In addition to ER and IR, passive and active forward flexion and abduction need to be recorded. Palpation of the shoulder during ROM can uncover crepitus, which may indicate bursal thickening, labral tear, or rotator cuff pathology. Subacromial impingement is assessed using the Hawkins-Kennedy and Neer tests.

Muscle strength should be evaluated after shoulder ROM is determined. Careful manual muscle strength testing should aim to isolate the muscle being tested and compare to the contralateral, uninjured side.

Several tests are helpful to examine the labrum of the throwing athlete. The O'Brien active compression test is frequently used to evaluate for SLAP tears.⁵⁷ Although the sensitivity has been reported at 100% with 97% to 99% specificity in the general population, the reliability markedly decreases in athletes (78% sensitivity, 11% specificity).⁵⁸ The passive distraction test is performed supine with the shoulder flexed to 150 degrees and the forearm is passively pronated to elicit increased biceps tension.⁵⁹ Deep pain or a pop with passive pronation is suggestive of a superior labral injury. Speeds test, commonly used for biceps pathology, has shown variable sensitivity and specificity in identifying SLAP tears in the literature.^{60,61} The Crank test evaluates the anterior glenoid labrum although the sensitivity ranges widely in the literature from 13% to 91%.^{62,63} The posterior labrum is evaluated with the labral shear, jerk, and Kim tests. Differentiating biceps pathology can be particularly challenging; however, the "3-pack" exam (O'Brien sign + throwing test + bicipital tunnel palpation) described by Taylor et al. has demonstrated excellent sensitivity in diagnosing both intra- and extra-articular injuries to the biceps labral complex.⁶⁴ The shoulder should also be evaluated for frank instability by testing the integrity of the anterior and posterior labrum as well as testing for a sulcus sign.

An important but often underemphasized part of the shoulder exam is evaluation of the scapulothoracic joint. Many injured throwers demonstrate loss of ER control and elevation and posterior tilt of the scapula, which are manifested as medial scapular border winging.⁶⁵ Both scapulas should be visualized with the arms at the sides; note is made of any asymmetry of tilt, rotation, elevation, or depression. As discussed, the SICK scapula presents with a static "dropped" appearance along with dynamic dyskinesis.¹⁵ Evaluation of scapular movement and scapulothoracic rhythm in comparison to the contralateral side is important. When evaluating dyskinesis, the scapular assistance test (assisted upward rotation and posterior tilt) and scapular

retraction test (manually placing the scapula in a retracted position) are diagnostic by improving pain and increasing motion and strength respectively.⁴²

In summary, examination of the painful throwing shoulder must be complete and systematic as these can be complex, subtle injuries. The examiner must develop a routine to avoid missing important findings. Additionally, the examiner must avoid focusing only on the shoulder and instead perform an assessment of the entire kinetic chain. The input of a physical therapist and/or athletic trainer well versed in overhead athletes also provides tremendous benefit.

IMAGING

Complete evaluation of the shoulder in an overhead throwing athlete should include the standard radiographic views of the shoulder consisting of anteroposterior, axillary, and outlet views. These images allow visualization of the glenohumeral articulation, as well as acromial morphology and visualization of the inferior glenoid. Other useful views may include the Stryker notch and West Point views depending on the situation.

After obtaining plain radiographs, the next step is generally magnetic resonance imaging (MRI). In overhead throwing athletes, the most common lesions are partial-thickness rotator cuff tears and glenoid labral pathology. MRI arthrograms (MRA) involve the use of intra-articular contrast material and can be helpful in delineating rotator cuff (particularly partial-thickness tears) and labral pathology with some studies demonstrating increased accuracy over plain MRI.^{66,67} However, the difference is not dramatic⁶⁷ so our bias has generally been against the routine use of MRA for the reason that it adds a relatively large volume of fluid into the joint and distorts normal anatomy and potentially distracts from other pathology. The use of MRI alone requires modern protocols on high-quality (3-T) scanners and partnering with a trusted set of radiologists; in doing so, we have found this to be far preferable. Beyond the traditional sequences, throwers often warrant placing the arm in the abducted ER position (ABER) to delineate more subtle labral and articular-sided rotator cuff pathology (Fig. 45.4).⁶⁸ In the setting of internal impingement pathology, MRI findings include articular-sided degeneration and tearing at the junction of the posterior supraspinatus and anterior infraspinatus tendons with associated degeneration and tearing of the posterosuperior labrum. Subcortical cysts and chondral lesions may also be seen in the posterosuperior glenoid and humerus as a result of repetitive impaction.

Given that pitching is a repetitive, supraphysiologic exercise, caution must also be used when interpreting MRI imaging on throwers as there is a high rate of asymptomatic pathology. Connor et al.⁶⁹ reported that 40% of dominant shoulders in asymptomatic overhead throwing athletes had partial- or full-thickness rotator cuff tears in their throwing arm compared with no tears in the nondominant shoulders. Similarly, Halbrecht et al.⁷⁰ demonstrated a 30% rate of labral tears and 40% cuff tendinosis with dominant-arm MRIs performed in the ABER position compared with the contralateral side. Because MRI findings are often abnormal, particularly in elite athletes, the

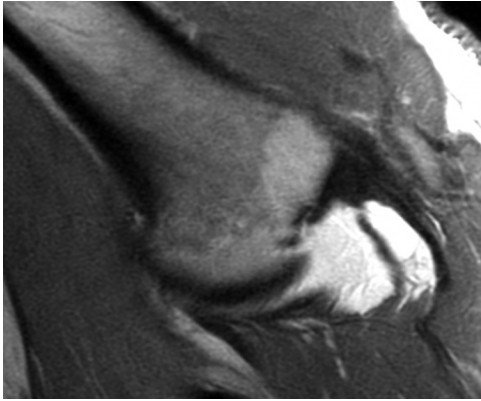


Fig. 45.4 A coronal image of a partial rotator cuff tear with the arm in the abducted and externally rotated position. Significant intrasubstance delamination of the tissue is seen in addition to a partial rotator cuff tear.

history and physical exam are critical to help delineate the source of the pathology and must be used in conjunction with imaging.

Computed tomography (CT) is less commonly used in the setting of the painful throwing shoulder. However, it remains the gold standard for evaluation of bony anatomy and measuring both humeral and glenoid version.⁷¹

DECISION-MAKING PRINCIPLES

Decision-making in the management of the throwing shoulder ideally follows an algorithmic approach based on available findings and evidence-based treatments. In general, the initial management of most shoulder injuries in overhead athletes is nonoperative, consisting of a period of rest, activity modification, and rehabilitation. Corrective efforts are directed not only at the shoulder but also at abnormalities of the entire kinetic chain. The younger the athlete, the more conservative our approach and the more willing we are to remove them completely from sports for a period of time. However, elite and professional athletes constitute a unique population in which relative rest and targeted therapy may be done to maintain fitness and certain skills while rehabbing. As discussed, the role of a thoughtful physical therapist who understands the demands on that specific athlete cannot be overstated, and communication between patient, physician, therapist, and team is critical.

Wilk and colleagues⁷² have described several phases of rehabilitation for overhead throwers with shoulder pain, including the acute phase, intermediate phase, strengthening phase, and return to throwing phase. In the acute phase, local therapeutic modalities focus on reducing pain and inflammation. Nonsteroidal antiinflammatory drugs, injections, ice, and iontophoresis may be used to accomplish this goal. A key to this phase is discontinuation of all activity to allow the shoulder to rest and the modalities to take effect. Shoulder motion must be addressed during this phase of rehabilitation. Stretches such as the sleeper's stretch (Fig. 45.5) are used to treat posterior capsule tightness. A major focus of the first phase of rehabilitation is improving the strength and muscle balance. Focus is therefore placed on



Fig. 45.5 In the sleeper's stretch, the patient lies on his or her side and stabilizes his or her scapula against the wall. Both the shoulder and the elbow are flexed 90 degrees. The nonaffected arm applies internal rotation to the affected arm. (From Kibler WB, Kuhn JE, Wilk K, et al. The disabled throwing shoulder: spectrum of pathology—10-year update. *Arthroscopy*. 2013;29:141–161.)

returning strength to the external rotators, scapular, and lower extremity muscles.

The goals of the intermediate phase are to progress with strengthening and enhanced flexibility of the shoulder. More aggressive isotonic strengthening activities are used as focus is placed on strengthening the external rotators, scapular retractors, protractors, and depressor muscles. Wilk prefers side-lying ER and prone rowing into ER. Strengthening of the lumbopelvic region and core are also a focus of the intermediate phase of rehabilitation. Jogging and sprinting are integrated into this phase as well to improve lower extremity strengthening and endurance. Upper extremity stretching exercises are also continued.

During the third phase of rehabilitation, the goals are to enhance power and endurance, perform functional drills, and gradually initiate throwing activities. Plyometric drills are also initiated during this phase. These drills are used to enhance dynamic stability, enhance proprioception, and gradually increase the functional stresses placed on the shoulder joint. Dynamic stabilization drills are also performed to enhance proprioception and neuromuscular control. During this phase, an interval throwing program is started. The interval throwing program is started once the athlete has a satisfactory clinical examination, nonpainful ROM, reliable kinetic chain movements, satisfactory isokinetic test results, and appropriate progress in his or her overall rehabilitation program.

Return to throwing is the fourth and final phase of the rehabilitation protocol. This phase typically involves progression of the throwing program. Although a number of approaches for this phase have been described with some variation, the general principles emphasize a strategic progression, often over a period of 6 to 12 weeks, during which the athlete progresses from the initial phase, consisting of short throwing over a limited duration to long toss and subsequently to position-specific throwing and/or throwing from the mound. Each step is taken in a careful

fashion, and any increase in symptoms prompts a short period of rest combined with a retreat to the prior phase to allow for resolution of symptoms. Once the athlete has accomplished the final phase of position-specific throwing, the decision can be made to return to competition. A judicious return to play (RTP) is recommended with careful monitoring of activity, innings, pitch count, and periods of rest aimed at limiting the risk of symptomatic recurrence.

Surgical options should be reserved for cases when nonoperative treatment fails and in the setting of certain acute injuries. The surgical procedure is determined by the shoulder pathology that is present; however, the approach in this population is patient-specific. All surgical pathology must be considered and managed systematically for optimal recovery of the athlete, which may require several concomitant surgical interventions such as SLAP repair and articular-sided rotator cuff débridement in the example of recalcitrant internal impingement.

TREATMENT OPTIONS

Aside from acute, operative injuries, the timing of surgery may depend on the relationship to the season of play. An athlete who is currently in-season may attempt to manage a structural shoulder injury conservatively to get through the season before undergoing surgery. This again emphasizes the importance of communication with all of the involved parties and allowing the athlete to participate in shared decision-making. How aggressive the timing and nature of the treatment are also depends on the age of the athlete and his or her level of play.

Internal Impingement

One of the challenges with internal impingement is differentiating normal adaptations from pathologic degeneration. We believe that many of the changes seen on imaging and even at the time of surgery are normal adaptations to the repetitive loads over time in throwers—in order to attain the supraphysiologic motions and forces necessary to throw, impingement must occur. Consequently, some level of change is also seen to the undersurface of the rotator cuff as well as the labrum, creating a so-called “thrower's footprint.” Parsing out typical impingement findings from a pathologic process is challenging and underscores the need for a comprehensive nonoperative regimen.

Internal impingement is associated with a spectrum of pathology and lends itself to multiple treatment options.⁷³ Those in the early states of impingement tend to have poorly localized pain and may have accompanying stiffness. Starting with a period of rest and nonsteroidal antiinflammatories is reasonable in this setting. For those throwers with more localized posterior shoulder pain, a prolonged period of rest (4 to 6 weeks) is in order along with formal physical therapy.³⁵ Therapy has been shown to be acutely therapeutic as well as protective against future injuries.^{5,15,28} The rehabilitation regimen is mainly directed at posterior capsular stretching, correcting scapular dyskinesis, and core strengthening. Given that one theory behind the pathogenesis of internal impingement is microinstability, anterior capsular stretching is generally avoided. Proper throwing mechanics should also be reviewed.

Only after clear failure of high-quality nonoperative treatment of appropriate duration should surgical intervention be considered. Throwers place tremendous demand on the shoulder and should generally give one pause prior to proceeding with surgery; however, internal impingement is a particularly challenging diagnosis to solve operatively.⁷⁴ Similar to the algorithm laid out by Heyworth and Williams,⁷⁵ we advocate for an approach dictated by the pathology. Exam under anesthesia (EUA) of both shoulders provides a sense of asymmetric laxity, which may be addressed with labral repair $\pm$ capsulorrhaphy. Rotator cuff tears in the setting of internal impingement are generally treated with débridement with the exception of high-grade tears, which will be discussed in the following section. Similarly, tears of the posterosuperior labrum can be débrided or repaired depending on the significance of the pathology.

Partial-Thickness Rotator Cuff Tears

The treatment of partial-thickness rotator cuff tears depends on several factors, including the depth and location of the tear, the quality of the tissue, and the athlete's age and playing position. Surgical options include débridement, tear completion and subsequent full-thickness repair, transtendinous repair, and intra-tendinous repair constructs. However, operative intervention on the rotator cuff of a thrower should be the absolute last resort. Our preference is for a comprehensive course of nonoperative management, and the athlete must definitively fail this conservative treatment by demonstrating persistent inability to throw prior to consideration for surgery.

In the general population, the conventional rule-of-thumb is that a tear involving less than 50% of the tendon insertion is débrided, whereas tears greater than 50% are more suitable for repair. These guidelines were initially adopted by several authors for overhead throwing athletes.^{76–79} However, Rudzki and Shaffer⁸⁰ contend that the demands of overhead athletes are different than the average individual and they have concerns about the integrity of a repair withstanding the high forces that will be placed on it. As such, they advocate for a 75% tear threshold prior to repair.

Articular-sided partial-thickness tears represent the most common rotator cuff injuries in throwers, and as such, the vast majority of our surgical management centers around simple débridement back to healthy, stable tendon. For the select few tears involving 80% to 90% thickness, we favor tear completion and anatomic repair. This general rule is particularly relevant for tears involving the anterior or posterior margins of the supraspinatus, which represent the attachment points of the rotator cable; we are less concerned about tears in the mid-portion of the supraspinatus.

Superior Labral From Anterior to Posterior Tears

Injury to the superior labrum in overhead throwing athletes can occur in conjunction with other pathologies or in isolation. Understanding the normal anatomic variants of the superior labrum and biceps insertion is critical to recognizing pathologic changes to the structures. Although the utilization of SLAP repair in the general population is decreasing, it still has a viable role in throwers. Similar to our treatment of partial-thickness rotator cuff tears, a thrower with a SLAP tear must fail extensive nonoperative

management and be unable to throw prior to consideration for surgery. In the setting of true pathology to the superior labrum, the stability of the SLAP tear dictates its treatment. Type I SLAP lesions are characterized by fraying of the central labrum without detachment of the biceps anchors and are treated with débridement. Type II lesions consist of isolated detachment of the superior labrum and biceps anchor either anteriorly (II-A), posteriorly (II-B), or involving the entire biceps anchor (II-C); these are typically treated with repair as the tear is believed to be unstable. A bucket-handle tear is present in type III lesions; these tears are treated with débridement and removal of the bucket-handle tissue. Type IV tears involve a bucket-handle component that extends up into the biceps tendon itself, and treatment involves repair of the SLAP ± biceps tenodesis as described in the next section.

Biceps Tenodesis

The popularity of biceps tenodesis is increasing dramatically⁸¹; however, there has been understandable trepidation to perform this procedure in overhead athletes.⁸² Generally, we have avoided tenodesis in throwers if at all possible given historically poor outcomes; however, in a very selective population, we have had success even with professional pitchers. Type IV SLAP tears that

involve a significant portion of the biceps tendon (>30%), such that it is unstable, are candidates. The other role for biceps tenodesis is in the setting of so-called biceps chondromalacia on the humeral head assuming this is consistent with history and physical exam as the etiology of the athlete's symptoms.^{51,83–85}

Posterior Capsular Contracture and Glenohumeral Internal Rotation Deficit

Posterior capsular contractures have generally responded well to conservative treatment. The “sleeper stretch” is typically the first-line treatment with historical failure rates being exceedingly low, especially in younger pitchers.^{15,28,53} There are some conflicting reports in more recent literature, though. A randomized, controlled trial of college-level overhead athletes found significant improvement of both IR and adduction with the sleeper stretch.⁸⁶ However, a systematic review attributed no benefit to the sleeper stretch and instead found cross-body adduction to be more efficacious.⁸⁷ Stretching certainly has limitations based on athlete compliance, and it can only be expected to overcome soft-tissue contracture with the understanding that certain bony adaptations may exist that inherently limit the motion of the shoulder. Still, nonoperative management is overwhelmingly the front-line treatment for posterior capsular tightness.



Authors' Preferred Technique

The surgical bar should be set very high when discussing the dominant shoulder of a thrower, and aside from acute traumatic injuries, we advocate that athletes complete comprehensive conservative management prior to operative intervention. It must also be emphasized that our discussion centers around general algorithms that should be individually tailored to the clinically relevant pathology of each athlete. When approaching surgery on a throwing athlete, it is important to understand the spectrum of injuries that are commonly seen, as outlined above, and to be prepared to address all intraoperative possibilities. We have found that utilizing a consistent team from nursing to surgical technician to anesthesiologist provides high-level, efficient care for this unique subset of patients.

Regional anesthesia provides tremendous benefit in shoulder surgery over traditional methods with regards to both intra- and postoperative courses⁸⁸; particularly with the advent of ultrasound guidance, we have found interscalene and supraclavicular blocks to be safe and effective in general practice. However, as it relates to the narrow population of elite and professional athletes, we generally avoid the use of neuraxial blocks given the risk, albeit small, of nerve injury and lasting dysfunction postoperatively. It is important to discuss this prior to surgery with the team as well as the patient to ensure a uniform understanding of the plan.

In terms of positioning, patients are placed in a modified beach chair that is fairly vertical such that the acromion is parallel to the floor. A beanbag is utilized and placed at the medial border of the scapula, which allows for maximal clearance as well as customized support and elevation of the ipsilateral trunk. An articulated arm holder is used for arm positioning throughout the procedure. Prophylactic antibiotics are always given. The skin is cleaned the evening prior and morning of surgery with chlorhexidine as well as formal surgical scrub prior to draping, also with chlorhexidine.

Internal Impingement

As discussed, internal impingement is a spectrum that envelops multiple injuries (Fig. 45.6), and treatment for each condition will be detailed in subsequent

sections. Internal impingement remains incompletely understood; however, we have found the theory of anterior micro-instability to be cogent at times. In our hands, anterior capsular procedures are only performed if we feel clear that instability is the primary source of pain as these patients will almost universally have a positive relocation test. One overriding principle in treatment is to take special care to avoid overtightening the anterior soft tissue envelope, which would limit the ER required to remain a competitive overhead thrower.

For anterior capsular plication, we prefer to capture capsule and labrum and repair it as one to the glenoid via knotless anchors instead of a capsule-to-capsule plication. While viewing posteriorly, we utilize a low anterior working portal directly above the superior border of the subscapularis tendon and a high

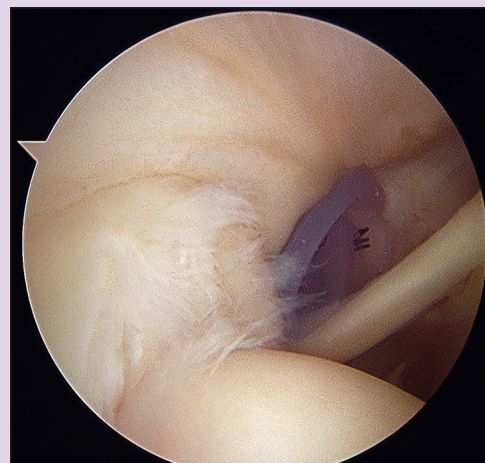


Fig. 45.6 An arthroscopic view from the posterior portal of an arm that is in abduction and external rotation. From this view, the rotator cuff can be seen encroaching on the superior labrum.

Authors' Preferred Technique—cont'd

anterolateral portal at the leading edge of the supraspinatus. Cannulas are placed in both portals for ease of suture passage and anchor placement. The arm is forward flexed and adducted over a large, sterile bump high in the axilla. This bump is a critical portion of the technique as it causes glenohumeral distraction and allows for excellent visualization and the necessary working space to pass instruments.

Many repair techniques are present in the literature, but our preference in throwers is unequivocally for a horizontal mattress configuration in knotless fashion. Both simple sutures and knots, even when tied away from the joint surface, have the potential to articulate with the humeral head given the extreme motions and forces that are applied to a thrower's shoulder. High-tensile strength, nonabsorbable #2 sutures are passed through the anterior capsulolabral tissue in horizontal mattress fashion with care to avoid capturing too much capsule as this pathology differs from gross, traumatic instability. Anchors are placed slightly onto the glenoid face to repair the tissue back to its anatomic position, but prior to final fixation, the arm is placed in a late-cocking position (ABER) to avoid overtightening. In our experience, 2-anchor fixation is generally sufficient for symptomatic anterior microinstability.

Rotator Cuff Tears

Following diagnostic arthroscopy, special attention is paid to the articular side of the cuff. If the pathology is clearly either low-grade or full-thickness, we then proceed with débridement or repair, respectively. But if the degree of tearing is not readily visualized, a spinal needle with a monofilament suture is passed through the tear via the subacromial space, and the area of concern is then visualized from the subacromial space.

The vast majority of rotator cuff tears in throwers are articular-sided and partial-thickness in nature (Fig. 45.7). As such, the bulk of our surgical management involves simple but thoughtful débridement with a shaver back to healthy, stable tendon. If at all possible, we favor débridement alone as the historical results of rotator cuff repair in pitchers have been poor.

For those tears involving more than 80% of the tendon thickness, we favor tear completion and restoration of the native anatomy via a lateralized footprint repair (Fig. 45.8).⁸⁹ Although intra- and transtendinous repair techniques exist, we have not found these to be predictable in overhead athletes. And while we believe that a broad footprint is important for repair, over-medialization negatively alters the native length–tension relationship, which can have devastating effects

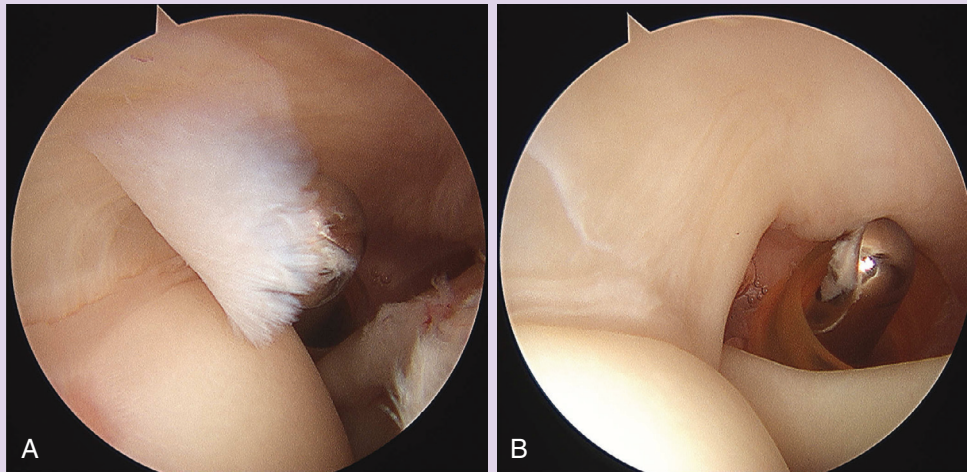


Fig. 45.7 (A) An arthroscopic view from the posterior portal looking at the articular side of the rotator cuff. A partial delaminated tear of the undersurface of the posterior rotator cuff is seen. (B) The frayed tissue is débrided, allowing evaluation of the true thickness of the tear.

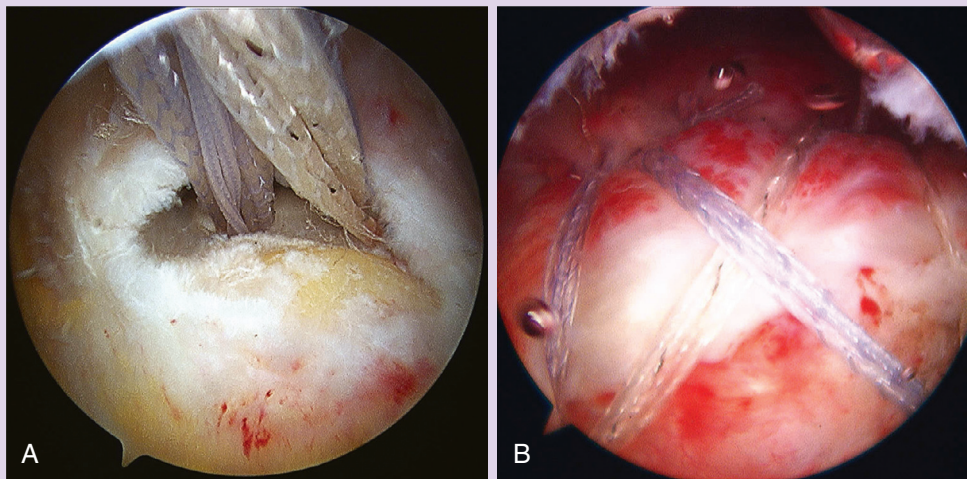


Fig. 45.8 (A) A crescent-shaped full-thickness tear of the supraspinatus with two double-loaded anchors placed at the articular margin. (B) Suture bridge repair of the full-thickness tear. Two medial and two lateral anchors were used for the repair.

Continued

Authors' Preferred Technique—cont'd

in a high-level thrower.^{90,91} The anatomy of the supraspinatus is readily visible and quite familiar to most sports surgeons as it begins at the articular margin and extends laterally. However, the infraspinatus has a more varied insertion as it begins superiorly at the articular margin but moves laterally as much as 16 mm at the inferior insertion.⁹² Precision in restoring a lateralized footprint is particularly relevant in an overhead throwing population where the goal is to recreate the "thrower's footprint" and avoid the potential for even small changes to motion or strength that can have amplified consequences in terms of function.

Prior to rotator cuff repair, a thorough subacromial bursectomy is performed to improve visualization. An acromioplasty is only performed if the presence of an impingement lesion is noted, which is uncommon in this population. The greater tuberosity is lightly abraded with a shaver in the region of the desired repair to enhance healing potential. A transosseus-equivalent, double-row construct is used in suture bridge fashion. Medial row anchor placement is critical in recreating native anatomy and preventing postoperative motion loss. The anterior supraspinatus anchor is placed immediately adjacent to the articular margin, which replicates the anatomic attachment of the rotator cable. Posteriorly, the infraspinatus anchor is placed 10 to 15 mm lateral to the articular margin. Medial row sutures are never tied as the knots can remain prominent and lead to irritation and a mechanical sensation during the throwing motion. The sutures are then brought down in crossing fashion to lateral row anchors just off the edge of the greater tuberosity. The number of anchors is dictated by the size of the tear although generally two medial and two lateral row anchors is sufficient in this population. We then critically assess the repair from both the subacromial and glenohumeral sides to ensure it restores the native footprint.

Superior Labral From Anterior to Posterior Tears

After diagnostic arthroscopy, particular attention is given to the "kissing lesion" (Fig. 45.9) between the posterosuperior labrum and undersurface of the posterior rotator cuff. While the circumferential labrum must be inspected carefully, the SLAP region should be of particular interest (Fig. 45.10). There are several normal anatomic variants, such as a sublabral foramen, Buford complex, and meniscoid labrum, which should be recognized and distinguished from true pathology. A close eye to hemorrhage or granulation tissue beneath the biceps-labral complex can help to distinguish pathology from a normal variant. The SLAP type and location must be identified as it will dictate accessory portal placement and ultimately treatment.

The type II-B (posterior) SLAP lesion is most common in throwers, so a portal of Wilmington is typically established under direct visualization using a spinal needle, and the trajectory of the needle should mimic that of the anchor. One

of the tenants of labral surgery that is often overlooked is adequate preparation of the labrum and the glenoid. While it is tempting to identify the tear and quickly begin the repair, inadequate preparation is a mistake that can compromise the healing potential of an already tenuous area. We take great care to fully mobilize the labrum with an arthroscopic elevator. This requires being slightly more medial on the glenoid but not undermining anteriorly to further destabilize the biceps anchor. Preparation of the glenoid bone bed deep is also critical to remove soft tissue and scar and provide a bleeding cancellous surface for healing; we favor the judicious use of shaver for this task. We will then assess the mobility of the labrum and its relation to the potential anchor site prior to proceeding with the repair.

High-tensile strength, nonabsorbable #2 sutures are passed through the posterior labrum in horizontal mattress fashion. Both suture limbs are grasped and placed over the template anchor position to ensure adequate position and tensioning. An anchor is then placed just barely onto the glenoid face, and the labrum is transfixed in knotless fashion. Most commonly, two anchors are required posterior to the biceps, and unless the tear truly extends anterior to the biceps anchor (type II-A/C), we avoid placing anchors anterior given concern for overconstraint. If there is anterior tear extension, the repair is conducted in identical fashion but with care to avoid the middle glenohumeral ligament.

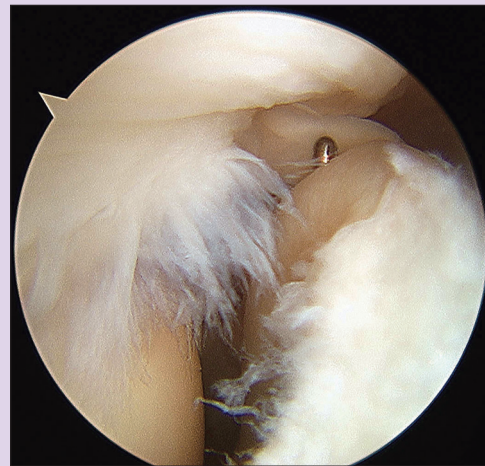


Fig. 45.9 Portals used for repair of a superior labral anterior to posterior tear. The high interval portal is commonly used.

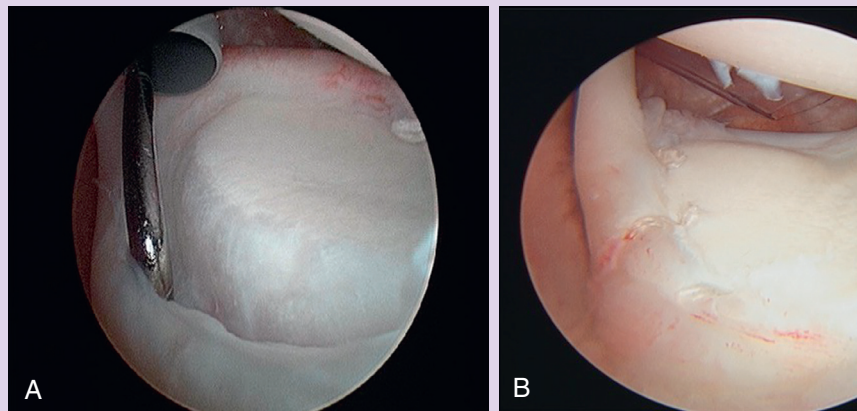


Fig. 45.10 (A) An arthroscopic view of a superior labral anterior to posterior tear. (B) This is repaired in knotless, horizontal mattress fashion.



Authors' Preferred Technique—cont'd

Biceps Tenodesis

Biceps tenodesis comes as a last resort in throwers, but our limited experience has been positive when adhering to the aforementioned indications. We advocate for a minimally invasive subpectoral technique with secure fixation through limited violation of the humeral diaphysis. Subpectoral tenodesis has the advantage of fully decompressing the intra- and extra-articular portions of the bicipital tunnel and allowing for cortically based fixation. The advent of a unicortical button has been a tremendous innovation as it improved cyclic loading over standard interference screw fixation, and the 3.2 mm drill hole minimizes the fracture risk for subpectoral tenodesis.⁸³

After arthroscopic tenotomy and addressing the necessary intra-articular pathology, we move to the open subpectoral portion. The beach chair position is altered from nearly vertical to closer to 45 degrees, and the arm is placed in slight abduction and ER to allow the surgeon to stand in the axilla. A small vertical incision is made just proximal to the axillary fold and centered over the inferior border of the pectoralis tendon. Our incision is more medial than sometimes described, and while it is closer to the brachial plexus, it allows for easy access to the humerus without requiring vigorous retraction on either the tendon or

medial soft tissue structures. After incising through skin, the clavipectoral fascia is split with a needle-tip bovie electrocautery, which provides tremendous precision. We then bluntly finger-dissect through the soft tissue and under the pectoralis tendon. A lateral retractor is placed momentarily to allow the biceps to be identified and pulled out of the wound; the tendon is then whip-stitched starting at the musculotendinous junction. The lateral retractor is again placed along with a gently held blunt retractor on the medial border of the humeral shaft. A 3.2 mm hole is drilled unicortically, and the tendon is secured to the diaphysis via a unicortical metal button. The suture limbs are then tied over top of the tendon for additional security.

The most important tips to avoid neurologic injury are to identify the correct incision and surgical plane and avoid prolonged or aggressive medial retraction. We have been very pleased with this technique in the general population as well as overhead athletes and have not had any fractures, which are a concern in throwers given the tremendous force and torque applied to the humerus. The most important principle, though, is not the technique but a very stringent patient selection process.

POSTOPERATIVE MANAGEMENT

A well-defined and thoughtful rehabilitation program is a critical component of a successful recovery and should be specific to the procedure performed. As is true preoperatively, a skilled physical therapist and athletic trainer play a vital role in the outcome as they are in much more frequent contact with the athlete, and open lines of communication between all parties are key. We have found that early cryotherapy in conjunction with scheduled oral antiinflammatory medications provides reasonable comfort and ability to participate in rehabilitation with only minimal narcotic usage. An exhaustive list of postoperative protocols is not practical, but we have listed below our approach to the most commonly addressed pathology in throwers. Our protocol follows five basic phases: acute postoperative, protection, intermediate stretching and strengthening, advanced stretching and strengthening, and RTP.

Following débridement of SLAP and rotator cuff tears, we initiate a fairly aggressive protocol (although the size and location of the tears must be considered). In the immediate postoperative period, we focus on controlling pain and inflammation while maintaining passive motion and scapular stability. We begin early active motion with progressive strengthening. Once the athlete demonstrates full motion and good manual muscle testing in all planes, he may focus on regaining and improving functional strength and normalizing scapular kinematics. Proprioceptive training and core, trunk, and lower extremity strength and endurance training are initiated. If all progresses well, the athlete is prepared to begin throwing in about 6 weeks. Our return to throwing protocol is detailed in [Tables 45.1 and 45.2](#).

In cases involving repair of the SLAP and rotator cuff and possibly capsular plication, the rehabilitation protocol is slowed significantly to allow for healing. The athlete is maintained in an abduction sling during phases 1 and 2 except while doing

therapy exercises. The goals of phase 1 (days 1 to 14) are to maintain the repair, minimize pain and inflammation, gradually increase passive motion, and practice dynamic stabilization to prevent muscular atrophy. In phase 2, we begin strengthening and proprioceptive drills between weeks 3 and 4 in those undergoing capsular or labral surgery; this is delayed until 6 weeks following rotator cuff repair. Phase 3 focuses on restoration of full passive motion, supine active motion, and reestablishing dynamic shoulder stability. As dynamic strength improves, the athlete enters phase 4, and our return to throwing protocol is initiated and continued for approximately 3 months. Overhead throwing typically begins at 5 to 6 months postoperatively.

The issue of RTP is addressed following completion of the throwing program. We typically utilize the Kerlan Jobe Orthopedic Clinic Shoulder and Elbow Score and expect a score greater than 85 for consideration of RTP. High-speed (240 frames per second) video analysis from both the front and side perspectives is also done for high-level throwers. Ultimately, the decision for final RTP must be made in conjunction with the player, trainer, and therapist.

RESULTS

Overall, the results following operative intervention on the shoulders of overhead athletes are sobering and should give surgeons pause when considering the treatment plan. A recent systematic review investigating collegiate and professional pitchers undergoing all forms of shoulder surgery found only a 68% RTP at 1 year, and while performance tended to improve from the preoperative state, most players did not achieve return to prior performance (RPP).⁹⁴ No doubt a major factor in this equation is the tremendous force across the glenohumeral joint and surrounding soft tissue structures in high-level throwers. Pitchers, in particular, require a finely tuned environment to function at an optimal level, and even small alterations to the balance within

TABLE 45.1 Interval Throwing Program

Distance	Number of Throws	Rest (Min)	Distance	Number of Throws	Rest (Min)
Phase 1 (50% Effort)			Phase 9 (75%)		
30'	20	5	90'	20	5
30'	20	10	90'	20	10
45'	10		120'	10	
Phase 2 (50%)			Phase 10 (75%)		
30'	20	5	90'	20	5
30'	20	10	90'	20	10
45'	20		120'	20	
Phase 3 (50%)			Phase 11 (100%)		
45'	20	5	120'	20	10
45'	20	10	120'	20	
60'	10		Phase 12 (100%)		
Phase 4 (50%)			120'	20	10
45'	20	5	120'	20	10
45'	20	10	120'	20	
60'	20		Phase 13* (Optional)		
Phase 5 (50%)			150'	20	5
60'	20	5	150'	20	
60'	20	10	Phase 14* (Optional)		
75'	10		150'	20	10
Phase 6 (50%)			150'	20	10
60'	20	5	150'	20	
60'	20	5	Phase 15* (Optional)		
75'	20		180'	20	10
Phase 7 (75%)			180'	20	
75'	20	5	Phase 16* (Optional)		
75'	20	10	180'	20	10
90'	10		180'	20	10
Phase 8 (75%)			180'	20	
75'	20	5			
75'	20	10			
90'	20				

*Our progressive program is designed to minimize re-injury risk and can be adapted for pitchers and position players. For phases 1–12, advance only 1 phase every 3 days with 2 days of rest following each workout. Beyond phase 12, ok to advance 1 phase every other day with 1 day of rest in between. Advancement is only allowed under pain-free conditions; dull pain and soreness often occur, however an athlete shoulder never throw through sharp pain. Of note, players should continue to work through their training program simultaneously to maintain flexibility, strength, and endurance.

the shoulder can precipitate a significant deterioration in function. Surgeons should allow these historical results to serve as a background for their approach and provide a basis for discussion with the athletes when setting expectations.

Rotator Cuff Débridement

Rotator cuff débridement remains our preference for partial-thickness tears of less than 80% thickness in part due to the poor results of repair; however, débridement is certainly not a guarantee to return to preinjury levels. A review of débridement in athletes demonstrated variable outcomes with success (as defined by the authors) ranging from 50% to 89% although

generally less favorable in high-level baseball players.⁹⁵ Payne et al.⁹⁶ provided an analysis of 43 athletes with partial, articular-sided rotator cuff tears treated with débridement and subacromial decompression. At 2 years, there was only a 66% satisfaction rate with 45% RPP in overhead athletes, which was notably lower than the 86% satisfaction rate and 64% RPP in their non-overhead counterparts. Subgroup analysis revealed that the athletes with the best results from débridement were those with isolated rotator cuff pathology without labral tearing, instability, or subacromial bursitis. Reynolds et al.⁹⁵ reported on 67 professional pitchers who underwent débridement alone for small partial-thickness tears with a high rate of RTP at 76% but only 55%

TABLE 45.2 Pitching Program

Distance	Number of Throws	Rest (Min)	Number of Throws	Rest (Min)
Phase 1: Winding Up Off Flat Ground (75%)			Phase 9	
60'6" (Warm-up throws)	20	5	Warm-up throws—20 fastballs at 50%	5
60'6"	20	5	20 Fastballs at 75%	5
60'6"	20		20 Fastballs at 75%	5
			20 Fastballs at 100%	5
			15 Fastballs at 100%	
Phase 2 (75%–100%)			Phase 10	
60'6" (Warm-up throws)	20	5	Warm-up throws—20 fastballs at 50%	10
60'6"	20	5	20 Fastballs at 100%	10
60'6"	20	5	15 Breaking balls and changeups	10
60'6"	20		20 Fastballs at 100%	10
			15 Fastballs at 100%	
Number of Throws			Phase 11	
Rest (Min)			Warm-up throws—20 fastballs at 75%	5
Phase 3: Off the Mound (100%)			20 Fastballs at 100%	10
Warm-up throws—20 fastballs at 50% intensity		5	6 Breaking balls and changeups	10
20 Fastballs at 50%			20 Fastballs at 100%, 6 breaking balls and changeups	
Phase 4			Phase 12	
Warm-up throws—20 fastballs at 50%		5	Warm-up throws—20 fastballs at 75%	5
20 Fastballs at 50%		5	20 Fastballs at 100%	10
20 Fastballs at 50%			10 Breaking balls and changeups	10
20 Fastballs at 50%			15 Fastballs at 100%	
Phase 5			10 Breaking balls and changeups	10
Warm-up throws—20 fastballs at 50%		5	15 Fastballs at 100%	
20 Fastballs at 50%		5	10 Breaking balls and changeups	10
20 Fastballs at 50%			15 Fastballs at 100%	
20 Fastballs at 50%			10 Breaking balls and changeups	
Phase 6			Phase 13	
Warm-up throws—20 fastballs at 50%		5	Warm-up throws—20 fastballs at 75%	
20 Fastballs at 50%		5	20 Breaking balls and changeups	
15 Fastballs at 75%		5	Batting practice—50–60 pitches	
15 Fastballs at 75%			Phase 14	
Phase 7			Warm-up throws—15 fastballs at 75%	
Warm-up throws—20 fastballs at 50%		5	15 Breaking balls and changeups	
20 Fastballs at 50%		5	Batting practice—70–80 pitches	
20 Fastballs at 75%		5	Phase 15	
20 Fastballs at 75%			Simulated game ^a	
20 Fastballs at 75%				
15 Fastballs at 100%				

^aThe simulated game is to be designed specifically for each individual pitcher.

Following phase 16 of the Interval Throwing Program, pitchers will progress to this program.

RPP. Maximum velocity also decreased from 94.2 to 90.1 miles/h. These results again underscore the difficulty obtaining predictable results when operating on elite throwers.

Rotator Cuff Repair

The results of rotator cuff repair have traditionally been poor in this population and are part of the driving force behind débridement for partial-thickness tears. Tibone et al.⁹⁷ conducted open acromioplasty and repair in overhead athletes with a 41%

RTP overall. RTP for pitchers was 40% but none were able to attain RPP. Mazoué and Andrews⁹⁸ reported on a small series of professional baseball players who underwent mini-open technique for full-thickness repair. Pitchers did poorly as only 8% were able to RTP; however, position players fared better with 75% RTP, particularly when the surgery involved their nondominant arm.

Although no direct comparison exists in overhead athletes between transtendinous repair and tear completion with

subsequent repair, a randomized trial in the general population did not demonstrate any statistical difference between the two techniques although there was a nonsignificant relative increase in power for the tear completion and repair group.⁹⁹ In a similar comparison, Shin¹⁰⁰ found equal satisfaction rating in each group but quicker recovery of motion and less pain in the initial post-operative period following repair after tear completion.

Ide⁵² initially reported the transtendinous technique in 6 throwers with an 83% RTP rate but only 40% RPP. Conway¹⁰¹ described his initial experience with intratendinous repair in 9 professional baseball players although this was a mixed cohort as all athletes had SLAP tears and half had anterior instability. Regardless, he demonstrated excellent results with 89% RPP at 1 year.

Our lateralized full-thickness repair in a series of 6 professional pitchers has allowed for 100% RTP and 83% RPP for at least one season.⁸⁹ We feel that this is the optimal technique and provides the highest likelihood of success in the face of a difficult injury. However, this high rate of return should be tempered by the fact that pitching productivity generally did not return to preinjury levels. Athletes played for an average of 3.3 seasons following surgery, but the mean number of innings pitched decreased from 1806.5 to 183.7 and performance (ERA) decreased slightly for two-thirds of players.

Superior Labral From Anterior to Posterior Tear Débridement

Early in the treatment of SLAP tears in throwers, suture anchors were not available, and thus débridement was the only option, but the results of unstable SLAP débridement were poor and deteriorated over time. Cordasco et al.¹⁰² evaluated 52 consecutive patients who underwent labral débridement and reported 78% excellent pain relief at 1 year, which decreased to 63% by 2 years. Only 45% of throwers whose SLAP tears were treated with débridement were able to return to their preinjury level of throwing. Altchek et al.¹⁰³ reported follow-up findings for 40 patients who underwent débridement for unstable SLAP tears. At an average follow-up of 43 months, 70% had moderate pain and 23% had severe pain. Only 1 out of 40 throwers was able to return to preinjury level of throwing. Alternatively, Tomlinson and Glousman¹⁰⁴ reported 75% of professional baseball players were able to RTP with mild or no pain following labral débridement; however, it should be noted their cohort included all labral tears, not specifically SLAPs. These results underscore the importance of an intact superior labrum and the need to carefully scrutinize the stability of the tear pattern.

Superior Labral From Anterior to Posterior Tear Repair

With improvement in technology and introduction of suture anchors, repair becomes a viable option. However, the results over the past 15 years have been inconsistent, with overhead athletes generally having the worst outcomes. Early on, Conway¹⁰¹ described an 89% RPP at 1 year following SLAP repair in a small cohort of baseball players. In a slightly larger series of type II SLAP repairs in overhead throwing athletes, Ide et al.¹⁰⁵ reported 90% RTP at 3 years. Of pitchers specifically, there was a 95% RTP rate but only 63% RPP.

Recent systematic reviews of type II SLAP repairs demonstrate positive but inconsistent results. One report denotes 88% good to excellent results in overhead athletes with more modern anchor repair but still only 63% RTP.¹⁰⁶ Another revealed good to excellent results ranging from 40% to 90% depending on the study.¹⁰⁷ They found that throwers had the most modest results, though, with 22% to 64% RTP rates in baseball players.

Finally, a recent case series of SLAP tears by Fedoriw et al.¹⁰⁸ involving a professional baseball organization highlights the difficult nature of this injury in throwers and particularly pitchers. Following a nonoperative algorithm, results were fairly similar with pitchers able to RTP at 40% with 22% RPP compared with 39% RTP and 26% RPP for position players. However, after SLAP repair, position players were much more likely to RTP at 85% with RPP of 54%, whereas pitchers had only a 48% RTP rate and 7% RPP.

Biceps Tenodesis

Although discussed, there is currently no literature on the results of biceps tenodesis in an isolated overhead athlete population.⁸²

COMPLICATIONS

Overall, shoulder arthroscopy is quite safe with a low incidence of complications. A recent analysis of 30-day morbidity of more than 9000 elective cases revealed a 0.99% complication rate with major morbidity of 0.54%.¹⁰⁹ The most common incident was return to the operating room (without specified reason) at 0.31%. Infection, albeit rare, did occur with 0.16% superficial and 0.01% deep rates. Thromboembolic events were equally uncommon with occurrence rates of 0.09% deep vein thrombosis and 0.06% pulmonary embolus. Only a 0.01% rate of peripheral nerve injury was noted. On multivariate analysis, risk factors included smoking history, chronic obstructive pulmonary disease, operative time greater than 1.5 hours, and American Society of Anesthesia class 3 to 4.

Fortunately, the overhead athletic population tends to be young and healthy and on the lower end of the risk spectrum. Care should always be taken to minimize complications, but fortunately, shoulder arthroscopy is a fairly safe procedure with minimal morbidity. From our perspective, though, the greatest risk is not postoperative complications as discussed above but rather failure to return to the supraphysiologic level of function required for throwing. Limitations in motion and persistent alterations in kinematics are always a possibility when performing any shoulder surgery on an overhead athlete, and even when seemingly minor, these changes can have dramatic effects on downstream function.

SUMMARY AND FUTURE CONSIDERATIONS

Overhead athletes are a high-demand but rewarding patient population, and physicians should approach their care with a solid knowledge base, realistic expectations, and stringent surgical criteria. Quite the opposite of elbow operations,¹¹⁰ the rate of shoulder surgery in throwers is decreasing both in our practice as well as nationally.^{111,112} As discussed, early postoperative outcomes for a variety of conditions have been modest, and while

results continue to improve, there should be trepidation prior to considering surgery. Conservative care should be the mainstay of treatment for most shoulder-related issues in this population with operative intervention truly being a last resort.

Several questions, such as the role for biceps tenodesis in throwers, remain unanswered but are certainly of interest around the world. The future of the field will likely focus on more specific treatment guidelines and operative indications, surgical techniques specific to this population, and evidence-based rehabilitation protocols and RTP guidelines.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Burkhart SS, Morgan CD, Kibler WB. The disabled throwing shoulder: spectrum of pathology part I: pathoanatomy and biomechanics. *Arthroscopy*. 2003;19(4):404–420.

Level of Evidence:

IV

Summary:

This three-part series of articles covers all aspects of the disabled shoulder in detailed fashion.

Citation:

Sewick A, Kelly JD, Rubin B. Physical examination of the overhead athlete's shoulder. *Sports Med Arthrosc Rev*. 2012;20(1):11–15.

Level of Evidence:

V

Summary:

Succinct guide to examining the shoulder of an overhead athlete.

Citation:

Kibler BW, Sciascia A, Wilkes T. Scapular dyskinesis and its relation to shoulder injury. *J Am Acad Orthop Surg*. 2012;20(6):364–372.

Level of Evidence:

V

Summary:

Scapular dyskinesis is a common clinical entity relating to the general population in addition to throwers, and it is often poorly recognized and not well understood. This provides a review of the salient points.

Citation:

Kibler WB, Thomas SJ. Pathomechanics of the throwing shoulder. *Sports Med Arthrosc Rev*. 2012;20(1):22–29.

Level of Evidence:

V

Summary:

Analysis of the pathology seen within the throwing shoulder. This provides a comprehensive evaluation of how throwing mechanics are altered in the setting of injury.

Citation:

Corpus KT, Camp CL, Dines DM, et al. Evaluation and treatment of internal impingement of the shoulder in overhead athletes. *World J Orthop*. 2016;7(12):776.

Level of Evidence:

IV

Summary:

A current review on the pathogenesis, evaluation, and treatment of internal impingement.

Citation:

Thorsness R, Alland JA, McCulloch CB, et al. Return to play after shoulder surgery in throwers. *Clin Sports Med*. 2016;35(4):563–575.

Level of Evidence:

V

Summary:

Specific focus on the data and challenges faced with getting throwers back to competition.

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